

Asad Ikram

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Prof. Dr. Sonja Utz
Everyday Media Lab
Leibniz-Institut für Wissensmedien (IWM)
Schleichstraße 6, 72076 Tübingen

Re: Researcher / PhD Position (m/f/d) in the Everyday Media Lab, reference 1014-2026

Dear Professor Utz,

The most consequential credibility judgment I have made about AI output was made at work, and I got it wrong. At ArtemisAI, the company I founded and where I am Chief Technology Officer, our dashboard reported an accuracy figure for the language classifiers I had built, and I trusted it as it climbed from 64.7 to 85.8 across five versions. Part of that climb was not the models improving. Up to the third version they were scored against a small set of human labels; from the fourth they were scored against labels produced by Claude, one of the two AI annotators whose output trains them, so the rising number increasingly measured agreement with another model rather than with people. The number looked authoritative because it was precise, it came from a system I understood, and it went up. I would like to spend my doctorate on exactly that kind of judgment: how people decide whether to believe information that AI has produced or filtered, and which cues they rely on when they do.

Why your lab. The Everyday Media Lab studies knowledge processes when people engage with social media or communicative AI, and encourages projects that connect the two, including the use of AI in workplace contexts. My own background sits on both sides of that connection. For seven years I have built the systems that collect what people post on social platforms, and more recently those that classify it: the social listening pipeline at M+C Saatchi Fluency that serves UK Government departments across ten platforms, and at ArtemisAI a pipeline in which AI models label social media content and smaller models learn from those labels. It runs in production ahead of our December launch. I know these systems from the inside. What I do not yet know is how people read what they produce, and that is the question your lab is built to answer.

A research question I would bring. My interest is in credibility judgments about AI-generated and AI-filtered information, in the two settings the lab names. In the workplace, professionals increasingly read AI-produced figures, summaries and confidence scores and act on them; I would ask which cues make such output believable, whether a precise number or a stated confidence raises trust independently of whether it is correct, and whether disclosing how a figure was produced changes the judgment. On social media, the same question applies to AI-generated content that people meet incidentally in their feeds. I would expect to answer these with controlled experiments that manipulate cues such as disclosure of AI origin or the presence of a confidence score, and I can add something less common to that design: large-scale social media data collected and classified at scale, to test whether effects seen in the lab hold in what people actually encounter.

My quantitative background. I hold an MSc in Data Analytics from the University of Portsmouth, awarded with Distinction in December 2025, with a project mark of 78. My dissertation was a quantitative empirical study of public reaction to UK economic policy: about 279,000 posts and comments collected from X, Reddit, YouTube and Quora under ethics approval, classified with a fine-tuned BERT model at macro-F1 0.878, validated against human coding at Cohen's kappa around 0.78, and tested with McNemar and bootstrap significance tests. Its central finding was that the same policy event produced structurally different sentiment and emotion profiles on different platforms. With my supervisor Dr Ella Haig I developed that into a methods manuscript that applies measurement invariance and score linking from psychometrics to machine-labelled text. It is complete but not yet submitted or published.

What I would need to learn, stated plainly. My master's is in data analytics rather than psychology or communication science. My quantitative empirical experience is real but observational rather than experimental: I have not yet designed or run a study with human participants, and I would come to the theories of media and social psychology as a learner rather than an expert. I have no publications. I do not speak German, and I would begin learning it on arrival. What I can offer in return is a strong quantitative and programming base, experience validating automated measures against human judgment, and first-hand knowledge of how the AI systems your participants would encounter are actually built.

I am available to relocate to Tübingen and to start in November 2026 or later. My CV, certificates and transcripts follow in this single file, as requested. Thank you for considering my application.

Yours sincerely,

Asad Ikram

Asad Ikram

MSc Data Analytics (Distinction) · Chevening Scholar 2024/25 · Applied machine learning and NLP, in production and in research
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EDUCATION

2024 - 2025 awarded Dec 2025	MSc Data Analytics, Distinction. University of Portsmouth, United Kingdom. Credit-weighted average 70.5/100 (Distinction threshold 70). Master's project 78/100. Supervisor: Dr Ella Haig. 180 UK credits, conventionally mapped to 90 ECTS. Modules included Intelligent Data and Text Analytics, Big Data and Machine Learning, Explainable AI, Data Management (80). Taught and assessed in English.
2017 - 2021 awarded Jan 2022	BSc Computer Science. National University of Computer and Emerging Sciences (FAST-NUCES), Pakistan. Four years full time, 130 credit hours completed. CGPA 2.75 on a 0.00 to 4.00 scale.

MACHINE LEARNING IN PRODUCTION

Feb 2025 - present	Founder and Chief Technology Officer , ArtemisAI Ltd (trading as ViralData), United Kingdom (remote). Running in production ahead of launch on 8 December 2026. • Six fine-tuned encoder classifiers for sentiment, emotion, toxicity, topic, language and intent, built from XLM-R, DistilBERT, DeBERTa-v3 and Detoxify bases and served from six SageMaker endpoints, called in parallel under 100ms. Custom per-sample loss weighted by label confidence, class balance and label source. • LLM labelling and distillation chain that produces the training data: Claude labels a stratified seed of about 6,000 posts together with its reasoning, the decision rules in that reasoning are mined into a prompt for DeepSeek, DeepSeek labels about 12,000 more, and any case below 0.80 confidence routes back to Claude. Result is about 18,000 machine-labelled items, which with synthetic cases makes a roughly 20,000-item confidence-weighted dataset, stratified 80/20. The models I fine-tune are encoder classifiers; I use Claude and DeepSeek through APIs and have not fine-tuned a generative model. • Deploy gate and retraining loop. A release ships only at macro-F1 above 0.85 on both a 97-post human holdout and a 4,000-item stratified test set, followed by a human check of 200 random posts. Weekly routing loop with three-tier confidence routing: roughly 75% auto-committed above 0.85, 18% sent to review between 0.60 and 0.85, 7% held below 0.60. Weekly KL-divergence drift alarms on output distributions and CI quality gates on every release. • Multimodal fusion classifier in production: a PyTorch network (1536 to 256 to 64 to 5, Adam, dropout 0.3, balanced class weights, best-validation checkpointing) over three concatenated 512-dimension CLIP embeddings of image, OCR text and caption, at macro-F1 0.93 on a hand-built gold set. Hybrid OCR pipeline at token-F1 0.928, Whisper speech transcription, and a networkx affinity graph applying network theory to a collaboration feature. • Data platform underneath the models: Meta Graph API ingestion of 30+ public pages daily into a layered lakehouse and Redshift warehouse, 111,000+ posts and 3.39 million comments processed. Products built on top: underperforming-post prediction, posting-time recommendation, drafting and crisis alerts. • <i>I also instrumented the measurement layer itself. The two annotation tiers agree on 82.9% of sentiment labels and 74.4% of emotion labels while the toxicity model reports 96.7% average confidence, and the dashboard's "accuracy" figure turned out to be inter-model agreement with no criterion behind it. Versions v1 to v3 were scored against the human holdout and v4 onward against Claude's labels, so part of the reported gain is the reference moving rather than the model improving. That moved my interest from building these instruments to asking what their outputs mean.</i>
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MACHINE LEARNING IN RESEARCH

Master's thesis: Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies. Fine-tuned a BERT classifier to **macro-F1 0.878** against BiLSTM and lexicon baselines, with human validation at Cohen's $\kappa \approx 0.78$, temperature-scaling calibration of the predicted probabilities, McNemar and bootstrap significance testing of model differences, chi-square topic tests, LIME and integrated gradients for interpretability, NRC emotion analysis linking anger and fear to negative policy reactions, and Prophet forecasting of the daily sentiment index (MAE ≈ 0.02). Collected ~279,000 public posts on UK economic policy from X/Twitter, Reddit, YouTube and Quora via official APIs and licensed feeds, under ethics approval (Portsmouth TETHIC-2025-111094) and GDPR-compliant procedures. *The finding that shaped my research interest was not the accuracy: the same policy event produced structurally different sentiment and emotion profiles on different platforms, which meant the cross-platform aggregate rested on an untested assumption, that the classifier measures the same construct everywhere it reads.*

RESEARCH OUTPUT

Ikram, A. and Haig, E. "Is the Platform Part of the Measurement? A Protocol and Simulation Study for Cross-Platform Equivalence of Machine-Labelled Policy Sentiment." **Manuscript, complete; not published and not submitted.** Target venue ICWSM 2027, Round 3 (15 January 2027). Treats each social platform as a survey mode and applies generalizability theory, measurement invariance and differential item functioning, and score linking to machine-labelled text. In a 500-replication simulation with known ground truth, an index that pools platform-level scores at face value reports the wrong sign of the aggregate on roughly a quarter of days, purely from measurement differences between platforms.

DATA ENGINEERING AND SOFTWARE

Dec 2024 - present	Data Engineer, consultant, M+C Saatchi Fluency, United Kingdom (remote) - Operate the multi-platform social listening pipeline behind campaign intelligence for UK Government departments and for Amazon, Ford, Nike and Reckitt: Sprinklr and Brandwatch feeds plus custom collectors into AWS (ECS Fargate, Step Functions, Glue, Athena), with SageMaker NLP sentiment models on the output. 68 collectors ingesting ten platforms daily into a governed 156-column master table; 6.7 million items in a six-week window at 99%+ uptime. Automated repair agent as a named CI workflow: when a source changes shape the system diagnoses the break, regenerates the extraction logic and tests it against live pages before shipping. - Anti-bot handling, proxy rotation and TLS fingerprint management; weekly KL-divergence drift alarms on pipeline output distributions; documented filter rules, removal lists and pre-delete backups on government-facing datasets.
Feb 2022 - Aug 2024	Senior Data Engineer, Dubizzle Labs (Dubizzle, Bayut, Zameen, OLX), Pakistan Led a three-engineer team owning 500+ regional collectors across 15 MENA countries on Kubernetes/EKS, gathering multilingual platform data in English, Arabic, Urdu and Hindi. Redesigned the layered warehouse (RDL, ODL, ADL), improving ETL/ELT throughput by 60%. Promoted twice in two years.
2018 - present	Earlier and concurrent engagements Fix.com, Canada, consultant, May 2023 to present (part time): Scrapy/Scrapyd pipelines processing 50M+ data points from 200+ sources; Great Expectations validation with automated schema-drift detection, cutting bad production data by 70%. VendueTech, Ireland, Lead Data Engineer, Jun to Jul 2025: led three engineers on real-time auction ingestion, migrated Azure to AWS. Prefe, Italy, consultant, Sep to Dec 2024: rebuilt legacy collectors, cutting failure rate 60%. CXG, Dubai, Dec 2025 to Jan 2026: production collectors for luxury retail at 99%+ completeness against advanced anti-bot protections. Freelance data engineer (Level 2 seller), Dec 2018 to Feb 2026: 300+ five-star projects across 400+ websites for 50+ international clients.

METHODS AND TOOLS

Machine learning	Transformer fine-tuning (BERT, XLM-R, DistilBERT, DeBERTa-v3, Detoxify); LLM-assisted labelling and distillation into smaller encoder models; confidence-weighted loss and class-imbalance handling; CLIP multimodal embeddings and fusion networks; Whisper transcription; OCR pipelines; BiLSTM; PyTorch; SageMaker training, endpoints and weekly retraining loops.
Evaluation	Macro-F1 against human holdouts and stratified test sets; inter-coder reliability (Cohen's κ); temperature-scaling calibration; McNemar and bootstrap testing; chi-square; KL-divergence drift detection; LIME and integrated gradients; Prophet forecasting.
Measurement	Generalizability theory, measurement invariance and differential item functioning, score linking; treating annotators and platforms as measurement instruments rather than as ground truth.
Data collection	Scrapy, Playwright, Selenium; official platform APIs (Meta Graph, Reddit, YouTube); licensed feeds; proxy and anti-blocking management; TLS fingerprinting; detection anatomy; agentic browsers.
Engineering	Python, PySpark, SQL; AWS (SageMaker, Fargate, Glue, Athena, Redshift, Step Functions, Lambda, DynamoDB); Airflow; Docker and Kubernetes; PostgreSQL; CI/CD with automated quality gates. R functional rather than fluent.

AWARDS, PUBLIC WORK AND LANGUAGES

Chevening	Chevening Scholar 2024/25, the UK Foreign, Commonwealth and Development Office's flagship global scholarship, selected from over 70,000 applicants across 160 countries and awarded to fund the MSc at Portsmouth.
Public work	Author and maintainer of web-scraping-guide.com , a free public engineering reference on web data collection: 24 sections covering 60+ libraries and tools, six anti-bot systems, detection anatomy, agentic browsers, legal and ethical constraints, production architecture and data quality.
Languages	Urdu (native). English: MSc taught and assessed in English, IELTS Academic 6.5 overall, March 2024. No German yet; I would begin learning on arrival.
Teaching	No formal university teaching. Mentored two junior engineers at Prefe and led three-engineer teams at Dubizzle Labs and VendueTech; wrote and maintain a public technical reference used by practitioners.

Only valid when viewed on port.ac.uk



UNIVERSITY OF PORTSMOUTH

Asad Ikram

has been awarded the degree of

**Master of Science in
Data Analytics**

with Distinction

having followed a programme of study approved by Academic Council

08 December 2025



Professor Graham Galbraith CBE,
Vice-Chancellor

Rebecca Di Pancrazio,
Academic Registrar



Student Number: 2291687 | Reference Number: 98611074-01-5ACA

Name of student **Asad Ikram**
Qualification **Master of Science in Data Analytics**
Award Classification **Distinction**
Date of Award **08 December 2025**
Mode of Study **Full Time**

University reference **2291687**
Date of birth **20 January 1998**
HESA reference **24100071550094239**

Awarding Institution **University of Portsmouth**
Teaching Institution **University of Portsmouth**

Record of Learning and Achievement

2024/2025	MSC DATA ANALYTICS - FHEQ Level 7	Mark	Credit	Decision
M32616	Masters Project	78	60	Pass
M33145	Programming for Data Analytics and AI	56	15	Pass
M33146	Big Data and Machine Learning Applications	67	30	Pass
M33147	Intelligent Data and Text Analytics	62	30	Pass
M33148	Data Visualisation and Exploratory Analytics	70	15	Pass
M33149	Data Management	80	15	Pass
M33516	Explainable AI	70	15	Pass
Credits gained for award			180	
Credits gained (ECTS)			90	

Information on the National Higher Education System can be obtained from National Academic Recognition Centre website:
www.enic-naric.net

For more information about this transcript please visit <https://myport.port.ac.uk/my-course/exams/after-your-assessments>

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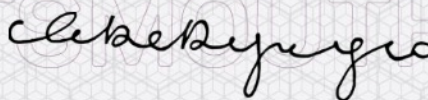
By email: awards@port.ac.uk

By telephone: +44 (0) 23 9284 7745

Official Signatures:



Professor Graham Galbraith CBE,
Vice-Chancellor



Rebecca Di Pancrazio,
Academic Registrar

Not valid unless authorised by signature

Date Transcript Printed: 20 August 2026

END OF TRANSCRIPT



*Denotes Pass/Fail Module

Student Number: 2291687 | Reference Number: 61386216-01-UUAE

National University of Computer & Emerging Sciences



This is to certify that
Mr. Asad Ikram
has been admitted to the degree of
Bachelor of Science (Computer Science)

With all the honours, privileges, and responsibilities pertaining thereto.

Awarded in the city of Islamabad on the Thirty First day of January in the year 2022.

A stylized, handwritten signature in black ink, consisting of a large, looped 'A' followed by a horizontal line and a small flourish.

Rector



A handwritten signature in black ink, reading 'Wasim Sajjad' in a cursive style.

Chancellor



NATIONAL UNIVERSITY

of Computer & Emerging Sciences
www.nu.edu.pk

Student Name: **Asad Ikram**
Date of Birth: **January 20, 1998**
Univ. Reg. No: **17K-3943**

Roll No: **17K-3943**
Degree: **BS (Computer Science)**

Fall 2017

Code	CourseName	Crd	Pnt	Grd	Rmk
CL 101	Introduction to Computing Lab	1	2.00	C	
CS 101	Introduction to Computing	3	1.00	D	
EE 182	Basic Electronics	3	2.00	C	
MT 101	Calculus - I	3	3.00	B	
SL 101	English Language Lab	1	3.00	B	
SS 101	English Language	3	3.00	B	
SS 111	Islamic and Religious Studies	3	3.00	B	

Credits Attempted: 17 GPA: 2.41
Credits Earned: 17 CGPA: 2.41

Fall 2018

Code	CourseName	Crd	Pnt	Grd	Rmk
CL 103	Computer Programming Lab	1	1.33	D+	R-1
CS 201	Data Structures	3	0.00	F	
CS 211	Discrete Structures	3	1.33	D+	
EE 213	Comp. Organization and Assembly Lang.	3		W	
EL 213	Comp. Organization and Assembly Lang. Lat	1		W	
MG 103	Finance and Accounting	3		W	
MT 104	Linear Algebra	3	2.33	C+	

Credits Attempted: 43 GPA: 1.23
Credits Earned: 40 CGPA: 2.15

Fall 2019

Code	CourseName	Crd	Pnt	Grd	Rmk
CL 203	Database Systems Lab	1	4.00	A	
CL 205	Operating Systems Lab	1	2.33	C+	
CS 203	Database Systems	3	3.33	B+	
CS 205	Operating Systems	3	3.67	A-	
CS 301	Theory of Automata	3	3.33	B+	
MT 206	Probability and Statistics	4	3.33	B+	

Credits Attempted: 62 GPA: 3.38
Credits Earned: 62 CGPA: 2.69

Fall 2020

Code	CourseName	Crd	Pnt	Grd	Rmk
CS 302	Design and Analysis of Algorithms	3	2.00	C	
CS 303	Software Engineering	3	3.33	B+	
CS 406	Web Programming	3	3.33	B+	
CS 422	Human Computer Interaction	3	3.33	B+	
EE 204	Computer Architecture	3	2.33	C+	
SS 118	Psychology	3	3.67	A-	

Credits Attempted: 97 GPA: 3.00
Credits Earned: 97 CGPA: 2.76

Fall 2021

Code	CourseName	Crd	Pnt	Grd	Rmk
CS 4036	Software Testing	3	2.33	C+	
CS 4038	Data Mining	3	1.00	D	
CS 4040	Remote Sensing and GIS	3	3.33	B+	
CS 4092	Final Year Project - II	3	4.00	A	
MG 1001	Fundamentals of Management	3	3.00	B	
MG 1002	Marketing Management	3	2.67	B-	

Credits Attempted: 130 GPA: 2.72
Credits Earned: 130 CGPA: 2.75

Spring 2018

Code	CourseName	Crd	Pnt	Grd	Rmk
CL 103	Computer Programming Lab	1	0.00	F	
CS 103	Computer Programming	3	1.33	D+	
EE 227	Digital Logic Design	3	1.67	C-	
EL 227	Digital Logic Design - Lab	1	3.00	B	
MT 115	Calculus - II	3	2.67	B-	
SS 113	Pakistan Studies	3	3.33	B+	
SS 122	English Composition	3	3.00	B	

Credits Attempted: 34 GPA: 2.29
Credits Earned: 33 CGPA: 2.35

Spring 2019

Code	CourseName	Crd	Pnt	Grd	Rmk
CS 201	Data Structures	3	3.00	B	R-1
EE 213	Comp. Organization and Assembly Lang.	3	4.00	A	
EL 213	Comp. Organization and Assembly Lang. Lat	1	2.67	B-	
MT 207	Numerical Methods	3		W	

Credits Attempted: 47 GPA: 3.38
Credits Earned: 47 CGPA: 2.47

Spring 2020

Code	CourseName	Crd	Pnt	Grd	Rmk
CL 309	Object Oriented Analysis and Design Lab	1		S	
CL 323	Computer Networks Lab	1		S	
CS 309	Object Oriented Analysis and Design	3		S	
CS 323	Computer Networks	3		S	
CS 401	Artificial Intelligence	3		S	
CS 402	Compiler Construction	3		S	
MG 103	Finance and Accounting	3		S	

Credits Attempted: 79 GPA: N/A
Credits Earned: 79 CGPA: 2.69

Spring 2021

Code	CourseName	Crd	Pnt	Grd	Rmk
CS 4104	Applied Machine Learning	3	3.00	B	
CS 449	Professional Issues in IT	3	2.33	C+	
CS 491	Project - I	3	2.67	B-	
MT 207	Numerical Methods	3	2.67	B-	
SS 108	Technical and Business Writing	3	3.00	B	

Credits Attempted: 112 GPA: 2.73
Credits Earned: 112 CGPA: 2.75

CGPA Required: 2.00
CGPA Earned: 2.75

Credits Required: 130

Degree Status: **Completed**

Credits Transferred: 0
Credits Earned: 130
Credits Completed: 130

April 01, 2022

NATIONAL UNIVERSITY
of Computer & Emerging Sciences
Islamabad

Controller Examinations

Test Report Form

ACADEMIC

NOTE Admission to undergraduate and post graduate courses should be based on the ACADEMIC Reading and Writing Modules. GENERAL TRAINING Reading and Writing Modules are **not** designed to test the full range of language skills required for academic purposes. It is recommended that the candidate's language ability as indicated in this Test Report Form be re-assessed **after two years** from the date of the test. To find out more about IELTS, IELTS band scores and the CEFR levels, please visit ielts.org/scores

Centre Number	PK011	Date	28/MAR/2024	Candidate Number	507881
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Candidate Details

Family Name	IKRAM
First Name(s)	ASAD
Candidate ID	3540423513455



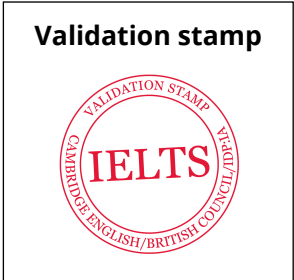
Date of Birth	20/01/1998	Sex (M/F)	M	Scheme Code	Private Candidate
Country or Region of Origin					
Country of Nationality	PAKISTAN				
First Language	URDU				

Test Results

Listening	7.0	Reading	6.5	Writing	6.0	Speaking	7.0	Overall Band Score	6.5	CEFR Level	B2
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Administrator Comments

Recognising organisations must verify this score at ielts.org/verify



Date	30/03/2024	Test Report Form Number	23PK507881IKRA011A
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